Multilingual Idiomaticity Detection and Sentence Embedding

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1 Abstract

Multiword expressions (MWEs) pose a persistent challenge to natural language processing (NLP) due to their frequent figurative usage and non-compositional semantics. In this paper, we address idiomaticity detection in a multilingual setting, building on the SemEval-2022 Subtask 2A framework. We focus on distinguishing idiomatic from literal usage under a zero-shot condition, where no target MWE in the dev set appears in training. The dataset used spans English and Portuguese with each MWE labeled as idiomatic (0) or literal (1). We first establish classical baselines with Logistic Regression and Naive Bayes on TF-IDF features, and a BiLSTM neural baseline trained on frozen FastText embeddings, which shows modest improvements but still lags behind transformer-based methods. Leveraging multilingual BERT, we replicate the official baseline and then introduce ModernBERT variants (MB.1 - MB.5) that progressively incorporate fine-tuning, data augmentation, custom preprocessing, and a custom cosine similarity loss to penalize high MWE-context similarity for idiomatic cases. Our experiments reveal that fine-tuning the BERT encoder yields substantial gains, with the final experiment achieving a macro-F1 of 0.692. The results underscore the efficacy of specialized training objectives and robust contextual embeddings for multilingual idiomaticity detection. Furthermore, these improvements have broader implications for downstream NLP tasks such as machine translation and sentiment analysis where accurately interpreting figurative language is crucial.

2 Introduction

The usage of Multiword Expressions (MWEs) is common throughout all languages. They are commonly created from cultural, societal, or special event-related occurrences. Their meaning is well established among native speakers and they are well integrated in everyday language usage (Ehren et al., 2020). Given the importance of MWEs, they are of particular interest in the field of natural language processing (NLP). Yet, they are shown to be a challenge due to the fact that the meaning of such expressions cannot be directly understood by their individual components (Choueka, 1988). To be more precise, MWEs can be used in a literal or in a more idiomatic way, which is understood by native speakers due to lexical cues and context. Thus, when it comes to idiomaticity classification, the classifier not only needs to be able to properly identify the MWE but also to be able to infer its meaning based on the given textual context. When looking at the example 'night owl', one might understand the reference of a person who is active late at night when used idiomatically or think of a nocturnal owl species when understanding it in a literal way (Tayyar Madabushi et al., 2022). This project aims to contribute to the goal of improving idiomaticity detection, especially including the cultural background introduced by the idiomatic expressions. This plays a crucial role across different use cases such as machine translation, parsing, and sentiment analysis (Omer and Hassani, 2025).

3 Related Work

Early work by Peng and Feldman (2016) framed idiom recognition as a distributional problem, proposing unsupervised methods based on inner products and scatter matrices in embedding space to distinguish literal from idiomatic usages. Tsvetkov and Wintner (2011) extended idiom detection by integrating multiple linguistic signals, morphology, syntax, and corpus statistics, within a Bayesian framework, demonstrating the benefit of feature fusion. More recently, Garcia et al. (2022) leveraged contextualized language models for idiomaticity detection, showing that transformer-based embed-

dings can capture subtle contextual cues when finetuned appropriately. Our approach builds on these insights by combining the contextual depth of transformer models with targeted finetuning and MWE aware input encoding. In doing so, it generalizes across MWE types and achieves stronger robustness in context-sensitive idiomaticity classification.

4 Task Formalization

The task of idiomaticity detection in Subtask A of SemEval-2022 is formally framed as a binary classification problem, where the goal is to determine whether a given multiword expression (MWE) within a sentence is used idiomatically or literally. At the core of this problem lies the non-compositional nature of idiomatic expressions (Titone and Connine, 1999), which renders traditional lexical or syntactic features insufficient for effective classification. This difficulty requires for a task formalization that makes use of contextual semantic cues to enable machine learning models to disambiguate idiomatic from literal usage which is enabled by creating a dataset of MWE expression usage including its context. This allows models that are able to process contextualized information to learn fine-grained differences.

Furthermore, the competition organizers, recognizing the difficulty of this task within the NLP domain, created two settings of the provided data with the goal of testing the generalisation ability of the trained models. In the Zero-Shot setting, models are tested on MWEs they have never seen during training, which emphasizes the necessity of pattern generalisation across contexts, rather than mere memorisation. The One-Shot setting introduces exactly one literal and one idiomatic example per target MWE, testing the model's ability to abstract a pattern from minimal data. Both settings highlight the importance of emergent patterns in usage and reinforce that idiomaticity can only be inferred when such patterns are grounded in contextual representation.

This formal framing is particularly fitting for transformer-based models, which, unlike static embeddings such as Word2Vec or GloVe, can dynamically encode the same expression differently depending on its context. As emphasized in prior work (Tayyar Madabushi et al., 2022), the inclusion of context allows the model to disentangle the semantic intent behind an MWE and assess its idiomaticity based on how closely its usage aligns with either compositional or non-compositional interpretations.

5 Data

The used dataset in the competition contains 8,683 labeled sentences spanning over three languages: English (5,352) making up for around 71%, Portuguese (2,555) 28%, and Galician only 1% (776) (these values spread out over train, dev, test and eval). Since only the labels for the training and development dataset are available, we used those for our project implementation. The ratio can be seen in table 1 as follows with a total of 5579 examples.

Split	**EN**	**PT**	**GL**
Train Zero-Shot	3327	1164	–
Train One-Shot	73	73	63
Test	466	273	–
Total	3866	1510	63

Table 1: Number of examples per language in each dataset split.

Each example includes a target sentence containing a multiword expression (MWE) with two adjacent contextual sentences. The dataset was built upon Tayyar Madabushi et al. (2021)'s work, with added Galician content. The provided labels indicate whether whether each MWE is used idiomatically (0) or literally (1) within the target sentence. The data was collected from internet sources by judges and language experts employing the categorization of compositionality levels with the goal of creating a even representation of all levels. Compositionality in multiword expressions refers to the degree to which the meaning of an expression can be derived from the combined meanings of its constituent words, as opposed to idiomatic expressions where the overall meaning cannot be predicted from its parts (Cordeiro et al., 2019).

As described in chapter 4, the data was provided in two settings - zero-shot and one-shot. In the zero-shot setting, models must classify previously unseen MWEs as the training data is disjoint from the development and test set. For one-shot evaluation, exactly one idiomatic and one compositional example for each MWE in data is provided, allowing models to learn from minimal examples. This setup is particularly challenging for Galician, which contains only 63 training instances, primar-

ily serving to evaluate cross-lingual transfer to a lower-resourced language.

Given the time constraints and scope limit within this project, we decided to concentrate on the zero-shot setting. Since Galician language data was only provided within the one-shot setting, we conducted our experiments only within the languages English and Portuguese.

Table 2 above illustrates the class distribution across the Zero-Shot training and test splits, separated by language. English has a higher sample count overall, and the idiomatic (label 0) class slightly outnumbers the literal (label 1) class in both languages for the training split. This imbalance is more noticeable in Portuguese. The test set remains relatively balanced, which is important for fair model evaluation.

Figure 1 demonstrates two examples extracted from the dataset where, for each MWE, we have at least one sentence where it was used idiomatically and one literally. The MWE cluster distribution in the zero-shot setting shows that noun clusters are predominantly present with 27.4% being noun-noun constructions, followed by noun-adjective representing almost one third and adjective-noun compounds making up 24%. Overall, nouns are contained in more than three quarters of the whole MWEs. Part-of-speech tagging was performed using language-specific spaCy models for English (en_core_web_sm) and Portuguese (pt_core_news_sm). Instances with missing MWE patterns were excluded from this analysis for clarity.

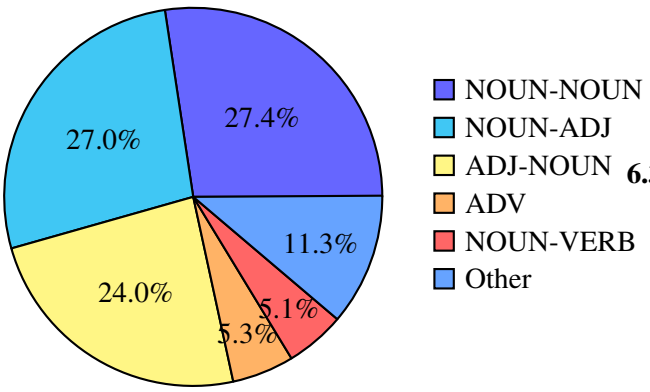


Figure 1: MWE Cluster Distribution of Zero-Shot Data

6 Experiments

This section outlines the models evaluated in our study, including baseline implementations and several BERT-based experiments.

6.1 Task Formulation

We frame idiomaticity detection as a binary classification problem. Given the input of sentences — preceding context, target sentence, and following context — the task is to determine whether a multiword expression (MWE) in the target sentence is used idiomatically or literally. All inputs are tokenized, encoded, and padded to ensure consistent input length across models. Training and evaluation are conducted using a held-out test set and final testing is done on the dev set. and standard binary classification metrics.

6.2 Baseline Models

We evaluated three baselines:

- BERT Baseline: We implemented the SemEval 2022 baseline using the author's script. It uses the multilingual BERT base model (bert-base-multilingual-cased), pretrained on Wikipedia in 104 languages using masked language modeling. A classifier head was trained on top of the frozen BERT encoder to perform binary classification.
- Naive Bayes, Logistic Regression: Trained on the target sentence using its TF-IDF representation. Preprocessing included stopword removal, lowercasing, lemmatization (spacy), vectorization (Tf-IDF).
- BiLSTM: Trained with pre-trained and fixed FastText embeddings. Preprocessing included punctuation, stopword, special characters removal and lowercasing. Hyperparameter training applied using grid search.

6.3 ModernBERT Experiments

- MB.1: Weights were frozen; only the classifier head was trained. No specific preprocessing was applied. Complete hyperparameter search.
- MB.2: Built on MB.1 - Finetune weights; train the classifier head. No preprocessing was applied. Hyperparameter search conducted.
- MB.3: Finetune weights; train attached classifier head. No preprocessing, but use augmented data. No hyperparameter search but used best hyperparameters from MB.2.

- MB.4: Finetune weights; train attached classifier, apply custom preprocessing. Use hyperparameters from MB.2 since performance in MB.3 was worse.

- MB.5: Finetune weights; train attached classifier, apply custom text preprocessing, use custom loss function (Cosine similarity of MWE and target sentence). Hyperparameter search only on similarity_alpha and learning rate, rest extracted from MB.2.

For BERT and BiLSTM, the input consisted of the previous, target (with MWE) and next sentence. For the classical ML approach, the target sentence was used. All models were implemented in Python using scikitlearn, PyTorch and Huggingface Transformers.

6.4 Model Implementation Details

modernBERT

The codebase includes scripts for training, evaluation, preprocessing, and configuration. All experiments were conducted using an NVIDIA Tesla V100-PCIE-16GB GPU. The structure and setup enable full reproducibility of results. Bayes Optimization was used for the hyperparameter search with was implemented using ray.tune. Sentences were tokenized using Huggingface's AutoTokenizer, padded to a fixed sequence length, and encoded with segment embeddings for the custom loss function. In the hyperparameter search the following parameters were tuned: warmup, learning rate, weight decay and alpha in MB.5 when the custom loss function was applied. The baseline multilingual BERT of the paper has 177854978 trainable parameters. The finetuned ModernBERT has 149606402 trainable parameters. The classifier head (MLP) of the ModernBERT has 768 neurons with GELU activation.

BiLSTM

A Bidirectional Long Short-Term Memory (BiLSTM) falls under the recurrent neural network approaches. Its powerful features is the capability of processing data in two directions (forward and backward) simultaneously which captures context in two directions. This makes it a powerful and popular choice, especially in the field of natural language processing (Rhanoui et al., 2019). Given the limited size of our dataset, we decided to train the classifier on a fixed pre-trained embedding as to not risk consequences like overfitting, instability or noise (Peters et al., 2019). We applied the English Fasttext embedding which is trained on 1 million word vectors trained on Wikipedia 2017 since it showed good performance results in a recently published paper (Arslan, 2024).

BiLSTM Model Architecture

With the goal of providing a stronger neural baseline beyond classical ML approaches, we implement a BiLSTM model in Keras. The input layer receives tokenized sequences of up to 147 tokens, which was sufficient to capture the target sentence plus optional context. These token indices are mapped to a 300-dimensional FastText embedding initialized from a pre-trained matrix, which remains frozen during training to mitigate overfitting on our relatively small dataset. As a result, the embedding layer typically accounts for around 2.7 million non-trainable parameters, while the rest of the network contains roughly 100k trainable parameters. Next, a bidirectional LSTM layer processes the sequence in forward and backward directions, each with a hidden size to be tuned. The output from the BiLSTM passes through a Dropout layer for regularization, then feeds into a Dense layer of 256 hidden units before the final Dense output of size 2 applies a softmax for binary classification (idiomatic vs. literal).

Hyperparameter Tuning and Final Configuration

We conducted a grid search over several parameters, including batch size (16 or 32), the number of LSTM hidden units (32 or 64), the dropout probability (0.2 or 0.3), and optimizer choice (Adam vs. RMSprop), training for up to 16 epochs. We employed early stopping in preliminary experiments when macro-F1 plateaued, striking a balance between avoiding overfitting and limiting training time. Each configuration was evaluated using macro-F1 on the dev set, an appropriate metric given our slight class imbalance. Our best results used 64 LSTM hidden units, a 0.3 dropout probability, batch size 16, and Adam at a 1e-3 learning rate.

Preprocessing Steps modernBERT

When it comes to preparing textual data, preprocessing steps for natural language preprocessing have a well-established integral guide of steps that are usually applied before training (Silva, 2021).

Recent work in natural language processing has challenged the long-standing assumption that standard preprocessing steps such as stopword removal or punctuation stripping are universally beneficial. In the context of transformer-based models like BERT, emerging evidence suggests that these steps may not only be unneccesary but could even impair model performance. For example, studies have shown that BERT is robust to variations in punctuation and capitalization and is capable of capturing nuanced contextual signals that traditional preprocessing might remove (Ek et al., 2020). Furthermore, recent work also uncovers that tokens always seen as irrelevant, like stopwords, articles and commas, and typically removed in the preprocessing step, in fact contribute significantly to long-range dependency modeling and contextual memory in transformer-based architecture (Razzhigaev et al., 2025). These findings supported our decision to minimize preprocessing in our BERT-based experiments, preserving as much of the original sentence structure and token information as possible. Rather, in experiment four and five, we decided to apply special tokens marking the boundaries for the sentences and for marking the MWE to analyse the impact.

Data Augmentation

Data augmentation is a commonly used technique in the NLP domain, especially when dealing with limited datasets. It includes approaches that artificially expand the training data with the goal of an improved generalization and robustness of the model (Şahin, 2021).

The paper "SemEval-2022 Task 2: Multilingual Idiomaticity Detection and Sentence Embedding" discusses the challenges associated with MWEs in NLP. It highlights that MWEs exist on a continuum of idiomaticity, meaning their interpretation can vary based on context.

We decided to apply the random swap data augmentation strategy, in which MWEs are kept fixed while in the target sentence words are randomly replaced. We hypothesized that it may either introduce unnatural context which could reduce the model's performance, or mitigate overfitting by promoting contextual generalization across diverse outputs. The goal was to encourage the classifier to develop contextual sensitivity, reduce overfitting to fixed expression structures and improve general robustness to both lexical and syntactic variability.

However, we encountered in the late stages of our experiments that the random swap was applied on each sentence to exchange at least one word (verb or noun), possibly including MWEs. This means that the lexical fixedness of idiomatic structures would be violated and the training data would introduce noise in the training process. Thus, we expect the third experiment to produce a significant reduction in performance.

Custom-Loss Function

When it comes to the approach of generating a custom loss function, we decided to pick linguistic features extracted from the data which then can be introduced in a class-weighted loss function. The hypothesis is that this would guide the encoder to capture idiomaticity more easily in its embedding space. We decided to apply the coside similarity calculation of MWE with its surrounding tokens in the target sentence. This approach is well supported by prior research (Liu and Hwa, 2018). Furthermore, it was also applied by the third ranked team in the competition in the zero shot setting. The hypothesis is that idiomatic MWEs often have lower semantic similarity between the MWE with their surrounding context compared to literal MWE usage. Thus, the custom loss function would penalize predictions harsher if they were idiomatic and had a high similarity and vice versa. We investigated the influence of a customized loss function.

Therefore, the two loss functions (**L1**, **L2**) are defined below.

L1 The `CrossEntropyLoss` provided by `Pytorch` was used as a base loss function. For the binary classification task at hand, a logit x and a target $y \in \{0, 1\}$, it is defined as

$$\ell(x, y) = -y \cdot \log\big(\sigma(x)\big) - (1-y) \cdot \log\big(1-\sigma(x)\big)$$

with

$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$

L2 For a given sample with token embeddings $\mathbf{h}_1, \ldots, \mathbf{h}_L$, let s_m and e_m denote the start and end indices of the MWE and s_t and e_t those of the target sentence. The average MWE embedding is computed as

$$\mathbf{e}_{\text{mwe}} = \frac{1}{e_m - s_m + 1} \sum_{i=s_m}^{e_m} \mathbf{h}_i,$$

and the context embedding (target sentence excluding the MWE) is given by

$$\mathbf{e}_{\text{context}} = \frac{1}{N} \sum_{i \in \{s_t, \ldots, e_t\} \setminus \{s_m, \ldots, e_m\}} \mathbf{h}_i,$$

where N is the number of context tokens. Their cosine similarity is defined as

$$\text{sim} = \frac{\langle \mathbf{e}_{\text{mwe}}, \mathbf{e}_{\text{context}} \rangle}{\|\mathbf{e}_{\text{mwe}}\| \|\mathbf{e}_{\text{context}}\|},$$

and with a label $y \in \{0, 1\}$ (with $y = 0$ indicating idiomatic usage), the similarity loss is given by

$$\mathcal{L}_{\text{sim}} = -\Big(y \cdot \text{sim} + (1 - y)(1 - \text{sim}) \Big),$$

so that the overall loss becomes

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \alpha \, \mathcal{L}_{\text{sim}}.$$

6.5 Evaluation Procedure

We evaluated all models using F1 (weighted and macro) score. Given that the evaluation or test set split was unavailable during the project phase, we decided to use the provided dev split (test.csv) for both validation during training and the final model evaluation. No further splitting of the training set was performed and all weight updates and optimization were performed on the training data. While this approach is agreeably a suboptimal approach, because the model indirectly optimizes towards the same date set used for evaluation, it prevents data leakage, since the training and dev split are disjoint from one another. This ensures that no MWE seen during training appears in the dev set, maintaining the zero-shot integrity of the task. That said, relying on the dev set for both validation and final scoring may lead to a slightly optimistic estimate of performance, as repeated use during tuning could bias the model toward better results on this particular set. While this does not violate the zero-shot condition (since MWEs are non-overlapping), it may not fully reflect generalization to a truly unseen holdout test set. Given the constraints of the task and data availability, we consider this a reasonable trade-off and followed this set up across all model runs.

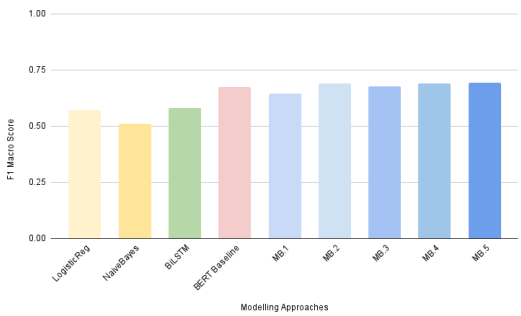
We compare the following model approaches:

- Logistic Regression and Naive Bayes: Classical ML baselines using TF-IDF features.
- BiLSTM: a neural baseline using frozen FastText embeddings and a bidirectional LSTM layer.
- SemEval BERT Baseline: the official baseline using a frozen multilingual BERT plus a classifier head.
- ModernBERT (MB.1-MB.5): a series of progressively refined BERT-based models, including data augmentation (MB.3), custom preprocessing (MB.4), and a custom similarity-based loss function (MB.5).

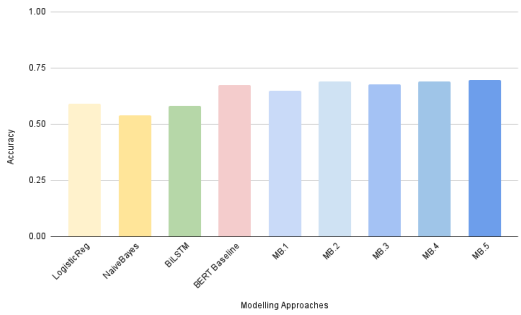
7 Results

Figure 2 presents the final metrics for each model with accuracy (a) and macro-F1 (b) score serving as our primary comparison metrics.

7.1 Overall Results



(a) Baseline Comparison Accuracy



(b) Baseline Comparison F1 Score

Figure 2: Baseline performance comparison using Accuracy and F1 Score.

The classical Logistic Regression baseline reaches about 0.59 accuracy and 0.57 macro-F1, which reflects strong precision for label 1 but low recall, ultimately producing a moderate outcome. Naive Bayes performs slightly worse at 0.54 accuracy and a macro-F1 of around 0.51, suggesting particular difficulty with more idiomatic usages. In

contrast, the BiLSTM shows a small boost at 0.58 for both accuracy and macro-F1, likely due to its ability to better capture context in a learned embedding space than TF-IDF-based method. It still lags behind the BERT-based approaches.

Continuing with the BERT-based approaches, we first note that the SemEval BERT Baseline stands at 0.674 accuracy and an F1 of 0.673 overall, demonstrating robust multilingual pretraining and generally outperforming the classical methods. To further explore potential gains, we trained a series of ModernBERT variants (MB.1–MB.5), the results are contained in table 11. In MB.1, which keeps BERT weights frozen, accuracy reaches 0.648 with an F1 of 0.675, already competitive with the BiLSTM. Moving to MB.2, where the BERT encoder is fully fine-tuned, performance jumps to 0.690 accuracy and 0.700 F1, surpassing the SemEval baseline and showing the clear benefits of end-to-end training. The subsequent experiments apply specific modifications: MB.3 adds data augmentation (0.678 accuracy, 0.675 F1), ultimately trailing MB.2 due to probable token corruption in the augmented data. MB.4 introduces custom preprocessing (0.690 accuracy, 0.675 F1), roughly matching MB.2. Finally, MB.5 pairs custom text preprocessing with a custom similarity-based loss, pushing performance further to 0.696 accuracy and 0.692 F1. These gains support the hypothesis that leveraging context-sensitive cues, alongside a specialized loss function for idiomatic usage, yields tangible improvements. Overall, MB.5 outperforms both the SemEval baseline and all classical models, highlighting the advantages of careful fine-tuning and tailored training objectives in multilingual idiomaticity detection.

7.2 Energy-Performance Trade-off

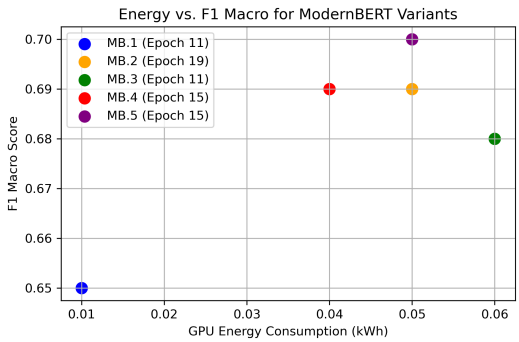


Figure 3: Energy Consumption vs. F1-Macro Score.

Figure 3 shows each ModernBERT variant (MB.1-MB.5) plotted by F1 Macro vs. GPU energy consumption (kWh). Experiment 1 requires only around 0.01 kWh but delivers the lowest score of 0.646. Meanwhile, Experiments 2 and 4 consume more energy yet reach 0.690 and 0.689. Experiment 3 contributed to the highest energy consumption with a score of 0.68 which shows that more epochs or heavier training does not guarantee higher performance if the data preprocessing or augmentation is flawed. Approach 5 achieves the highest score of 0.695 at a moderate cost (0.06 kWh). Overall, we see a slight upward trend in score with higher GPU usage, suggesting that longer or more intensive training yields marginal performance gains in idiomaticity detection.

7.3 Linguistic Error Analysis

For each experiment (MB.3–MB.5), we sampled 100 examples per category (TP, TN, FP, FN) to enable a balanced, qualitative error analysis. As Table 13 reflects, this sampling does not represent overall model performance but instead allows a controlled study of linguistic patterns, structural challenges, and model-specific tendencies in MWE classification, focusing on the topics MWE fixedness and cosine similarity.

Our aim was to answer the following hypotheses:

- HP1 for MB3: Random swap within each sentence, possibly including the MWE, will result in performance loss.
- HP2 for MB4: Does the custom preprocessing, MWE marking and separator insertion, improve the model's ability to correctly classify syntactically fixed idioms?
- HP 3 for MB5: Does the cosine similarity-based loss function in MB.5 help reduce false negatives for MWEs that are semantically fixed with their surrounding context?

Given that we already concluded the erroneous implementation of data augmentation in experiment 3, we expected the performance to drop which is shown in the metrics.

7.4 MWE Fixedness

MWEs can range in the degree of fixedness ranging from entirely rigid to highly flexible. (Alves et al., 2024). Rigid MWEs (e.g., "kick the bucket") are those whose form is almost entirely fixed, both

in terms of lexical items and syntactic arrangement, and which often exhibit complete idiomaticity. In contrast, flexible MWEs (e.g., "spilled the beans") allow for certain morphosyntactic variations in their components, though the extent and nature of this variation are not always easily predictable (Masini, 2019).

In figure 14 five manually extracted examples are visible which either fall under the fixed or flexible category. We then analyzed the FN/FP pattern across experiment MB4 and MB5 as seen in tables 15, 16.

MB4 consistently shows fewer false negatives (FN) and better precision on rigid MWEs. MB5 struggles especially with idioms being misclassified as literal (e.g., 8 FNs for panda car). For flexible MWEs, both models struggle, but MB5 occasionally outperforms MB4 (e.g., mental disorder), indicating less benefit from syntactic marking.

7.5 Cosine Similarity

In order to answer Hypothesis 3, we calculated the cosine similarity between the MWE and the sentence on the extracted predictions. Figure 5 shows that literal MWEs tend to have slightly higher cosine similarity scores than idiomatic ones, suggesting that literal expressions align more closely with their surrounding context. Figure 6 further supports this by showing that literal MWEs are more prevalent in the mid and high similarity bins, while idiomatic ones are relatively more frequent in the low similarity bin. Finally, Figure 7 breaks down classification outcomes by similarity bin and reveals that most correct predictions (TP and TN) occur in the mid range. In contrast, low similarity bins are associated with more misclassifications, particularly false negatives, idiomatic expressions being mistaken for literal ones. These findings suggest that cosine similarity can serve as a weak signal for idiomaticity, with idioms often being less semantically aligned with their surrounding context.

7.6 Result Discussion

Overall, fine-tuning BERT (MB.2 onward) surpasses classical ML and BiLSTM baselines. Notably, MB.5's custom similarity loss function yields a small yet consistent boost in macro-F1, supporting the hypothesis 3 (HP3).

The results indicate support for Hypotesis 2.MB4's custom preprocessing shows to improve classification accuracy on syntactically rigid idioms, as shown by the notable reduction in false negatives (e.g., panda car, flower child). These structural cues help the model retain idiomatic boundaries that are typically lost in plain-text input. In contrast, MB.5 tends to confuse such idioms for literal phrases. However, the improvement is notably structure-dependent: more flexible MWEs continue to pose challenges, indicating that the preprocessing gains do not generalize across the fixedness continuum.

At the same time, data augmentation (MB.3) offered no benefit, possibly due to the risk of token corruption. Future work should include extensive refinement of augmentation to avoid altering MWE tokens.

In summary, MB.5 emerges as the best performing approach for idiomatic detection in the zero-shot scenario, underscoring the value of specialized losses in bridging compositional and idiomatic usage.

8 Conclusion

Our study explored multilingual idiomaticity detection under a zero-shot setting, comparing classical ML baselines (Logistic Regression, Naive Bayes), a BiLSTM neural baseline, and multiple variants of a multilingual BERT model. While the classical baselines performances were adequate, they showed limited ability to handle the subtle contextual cues that often distinguish idiomatic from literal usages. The BiLSTM's capacity to learn context-aware embeddings offered moderate improvements, yet it still fell short of the performance of BERT-based approaches. Among the BERT methods, fine-tuning the encoder rather than freezing it yielded a consistent boost, validating the importance of adapting model parameters to our specific idiomaticity dataset. Although data augmentation attempts (MB.3) introduced token corruption for MWEs, undermining performance, nonetheless this result underscores the value of careful augmentation strategies that preserve idiom integrity. Our custom similarity-based loss function (MB.5) implementation further advanced the macro-F1, suggesting that idiomatic expressions often diverge semantically from their neighboring tokens, and explicit modeling of this gap is beneficial. Moreover, error analysis identified patterns in false positives and negatives, revealing how cultural or lexical complexity leads to misclassifications.

Overall, our findings show that fine-tuned transformer models consistently outperform classical baselines, with the MB.5 variant achieving the highest macro-F1. These results highlight the potential of specialized training strategies for figurative language detection, with implications of better handling of tasks like machine translation and sentiment analysis.

9 Limitations and Ethical Considerations

This study is subject to several limitations. First, due to data availability constraints, the development set was used for both, hyperparameter tuning and the final evaluation which might have introduced an optimistic bias in the reported performance metrics. Furthermore, due to the limited capacity within this project bound, the qualitative analysis is based on a small fixed sample size and may not fully capture the degree of errors. Furthermore, due to the limited scope of this project, we decided to focus the linguistic analysis on English.

Ethical considerations play a crucial role in idiomaticity detection because idioms are deeply rooted in a country's culture and often carry historical and emotional connotations. Widespread misclassification of such expressions by machine learning models could reinforce cultural misunderstandings and biases. Although the bulk of available data is in English, a growing body of academic work in other languages underscores the need to extend research to diverse linguistic contexts. In light of rapid advancements in language processing, it is essential to develop models that accurately represent a wide range of cultures and languages while continuing to work towards minimizing bias.

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Appendix A

A Model Implementation Details

BiLSTM Implementation: The requirements file for the BiLSMT implementation was created within a conda environment.

Best hyperparameters of ModernBERT Experiments:

- **Experiment 1:**
  - η: 4.97e^{-5}
  - λ: 1.28e^{-2}
  - ρ: 2.38e^{-1}
- **Experiment 2:**
  - η: 6.03e^{-5}
  - λ: 1.64e^{-2}
  - ρ: 1.31e^{-1}
- **Experiment 3:**
  - η: 6.03e^{-5}
  - λ: 1.64e^{-2}
  - ρ: 1.31e^{-1}
- **Experiment 4:**
  - η: 6.03e^{-5}
  - λ: 1.64e^{-2}
  - ρ: 1.31e^{-1}
- **Experiment 5:**
  - η: 7.35e^{-5}
  - λ: 1.64e^{-2}
  - ρ: 1.31e^{-1}
  - α: 3.39e^{-1}

Extra Information

Figure 1: The other category in contains the following clusters - PROPN-NOUN 3.7%, VERB-NOUN 3.0%, PROPN-PROPN 2.4%, ADJ-ADJ 2.0%, NOUN 0.2%.

Cosine similarity on the extracted predictions were calculated language specific with spacy's "en_core_web_md" and "pt_core_news_md", basic preprocessing was done. The low, mid and high bins were set as following: bins=[0, 0.4, 0.7, 1.0].

Appendix B

B Contributions and Learnings

Learning Objectives and Gained Learnings

This project has significantly supported my learning objectives by allowing me to deepen my understanding of both classical and modern NLP techniques. Having previously worked more with linguistic feature engineering, I was initially keen to continue in that direction. However, the focus on idiomaticity—specifically, its non-compositional and context-sensitive nature—presented an exciting and complex challenge. While our core implementation relied on a BERT-based model, which we identified early on as well-suited for capturing nuanced contextual embeddings, I explored the performance of BiLSTM architectures. Motivated by the fact that BiLSTMs are the predecessor of the current-state-of-the-art approaches, I wanted to explore how it would perform in contrast to our BERT implementation - this helped me learn more about deep learning approaches. Additionally, collaborating closely with the team, troubleshooting implementation issues, and iterating on the BERT model pipeline provided a steep learning curve — especially as this was my first direct experience working with transformer-based models. It was extremely interesting working with Docker, overall I have learned a lot conceptually and also by application in terms of good practices. Furthermore, having worked on implementing the cosine similarity custom function, I enjoyed a lot refreshing previously learned topics while applying this implementation into the use case of idiomaticity detetection. The discussions with the team were extremely fruitful.

Personal Contributions

In this project, I worked on exploring a custom cosine similarity loss, which was inspired by the third-best ranking team in the SemEval 2022 competition in the zero-shot setting (NER4ID). This involved the insertion of special separator tokens to effectively isolate MWEs and sentence boundaries which was necessary to extract the contextual embeddings from BERT's final hidden layer. These

embeddings were then integrated into the cross-entropy loss calculation implementation.

In addition, I implemented the baseline BiLSTM model, enabling a comparative analysis between a classical deep learning architecture and our transformer-based methodology. This allowed us to assess model performances across different methodologies. Beyond this, I conducted a detailed linguistic analysis aimed at uncovering meaningful patterns within our results. Although only to a limited extend, I first manually extracted MWEs with the theoretical backing of "fixedness vs flexible" structure. Then I assessed the performance in FN, FP across experiment 4 and 5 keeping these examples in main focus. Furthermore, I performed descriptive data analysis on the extracted predictions and finally focused on assessing the cosine similarity in correlation to the class labels, aiming to extract either supporting or conflicting evidence regarding our hypotheses.

Main take-away

Main findings from this project include: (1) Fine-tuned transformer models outperform classical and earlier neural baselines in zero-shot idiomatic detection; (2) Specialized preprocessing or loss functions can provide measurable gains, but must be handled carefully to avoid data corruption; and (3) Unsupervised or limited-resource strategies, such as data augmentation, demand thorough validation to prevent inadvertently harming performance.

In future work, we can aim to refine our augmentation methods to better preserve MWE structure, exploring other possible practices. A different direction to explore in future work would be to dive deeper into extracting linguistic features that better translate the phenomenon of MWE into machine readable patterns. Another interesting area to explore would be to work on extending the language corpus availability and to introduce specific domains within MWE usage to assess how our model would generalize.

Appendix C

C Appendix

Table 2: Logistic Regression Evaluation Report

Label	Precision	Recall	F1-Score	Support
0	0.53	0.91	0.67	336
1	0.81	0.33	0.47	403
Accuracy		0.59		
Macro Avg	0.67	0.62	0.57	739
Weighted Avg	0.68	0.59	0.56	739

Table 3: Naive Bayes Evaluation Report

Label	Precision	Recall	F1-Score	Support
0	0.48	0.70	0.57	336
1	0.60	0.37	0.46	403
Accuracy		0.54		
Macro Avg	0.54	0.53	0.51	739
Weighted Avg	0.54	0.52	0.51	739

Table 4: BiLSTM Evaluation Report

	Precision	Recall	F1-score	Support
Class 0	0.53	0.73	0.61	335
Class 1	0.67	0.46	0.54	403
Accuracy			0.58	738
Macro Avg	0.60	0.59	0.58	738
Weighted Avg	0.60	0.58	0.57	738

Table 5: SemEval BERT Baseline Evaluation Metrics

Metric	Value
Overall Performance	
Evaluation Loss	2.082
Accuracy	0.674
F1	0.673
Evaluation Stats	
Runtime	3.45 seconds
Samples	739
Samples/Second	214.116
Steps/Second	26.946
Zero-Shot Performance	
Zero-Shot EN	0.662
Zero-Shot PT	0.639
Zero-Shot EN,PT	0.673

Table 6: Experiment MB.1 Evaluation Metrics

Metric	Value
Overall Performance	
Evaluation Loss	0.635
Accuracy	0.648
Balanced Accuracy	0.661
F-Scores	
F1	0.675
F1 Macro	0.646
F1 Weighted	0.643
Precision/Recall	
Precision	0.582
Recall	0.804

Table 7: Experiment MB.2 Evaluation Metrics

Metric	Value
Overall Performance	
Evaluation Loss	2.848
Accuracy	0.690
Balanced Accuracy	0.699
F-Scores	
F1	0.700
F1 Macro	0.690
F1 Weighted	0.689
Precision/Recall	
Precision	0.625
Recall	0.795

Table 8: Experiment MB.3 Evaluation Metrics

Metric	Value
Overall Performance	
Evaluation Loss	2.969
Accuracy	0.678
Balanced Accuracy	0.683
F-Scores	
F1	0.675
F1 Macro	0.678
F1 Weighted	0.678
Precision/Recall	
Precision	0.624
Recall	0.735

Table 9: Experiment MB.4 Evaluation Metrics

Metric	Value
Overall Performance	
Evaluation Loss	3.138
Accuracy	0.690
Balanced Accuracy	0.692
F-Scores	
F1	0.675
F1 Macro	0.689
F1 Weighted	0.691
Precision/Recall	
Precision	0.645
Recall	0.708

Table 10: Experiment MB.5 Evaluation Metrics

Metric	Value
Overall Performance	
Evaluation Loss	2.667
Accuracy	0.696
Balanced Accuracy	0.700
F-Scores	
F1	0.692
F1 Macro	0.695
F1 Weighted	0.696
Precision/Recall	
Precision	0.641
Recall	0.753

Experiment Group	**Model**	**Accuracy**	**F1**$_{macro}$
Reproduction	SemEval Baseline	0.674	0.673
Naive Baselines	Logistic Regression	0.590	0.570
	Naive Bayes	0.520	0.510
	BiLSTM	0.580	0.580
ModernBERT	Experiment 1	0.650	0.646
	Experiment 2	0.690	0.690
	Experiment 3	0.678	0.678
	Experiment 4	0.690	0.689
	Experiment 5	**0.696**	**0.695**

Table 11: Comparison of the provided SemEval baseline, trained baseline models and the best models obtained from the hyperparameter search within the ModernBERT experiments.

Experiment	False Negatives	False Positives	True Negatives	True Positives
MB3	100	89	100	100
MB4	100	97	100	100
MB5	100	84	100	100

Table 12: Error Type Distribution Across Experiments

Category	Example and Explanation
Rigid/Fixed MWEs	
panda car	"The police arrived quickly in a panda car." Refers to a small UK police patrol vehicle. Term is fixed; replacements like "bear car" or "police vehicle" lose the idiomatic meaning.
bow tie	"...bow tie cinemas will also open on Feb. 5 at its Ultimate Annapolis Mall..." Used literally as a compound noun or part of a brand. Meaning is compositional but rephrasing is rare, making it moderately rigid.
chain reaction	"One error led to a chain reaction of failures across the system." Has fixed idiomatic/metaphorical use. Reordering or replacing parts distorts recognizability.
flower child	"He was a true flower child of the 60s, embracing peace and free love." Fixed cultural expression; altering any word distorts the meaning.
health check	"The server went down during a routine health check." Used in both tech and medical contexts. Phrase is fixed; substitutions lose clarity.
Flexible MWEs	
economic aid	"The country received significant economic aid from international organizations." Highly flexible; can be reworded as "economic support", "financial aid", etc. without losing meaning.
insurance company	"...a domestic insurance company in the United Kingdom, is an alien insurance company in the U.S." Allows rephrasing (e.g., insurer, insurance provider); non-idiomatic and syntactically flexible.
mental disorder	"She was diagnosed with a severe mental disorder." Allows variation (e.g., "disorder of the mind", "psychological disorder"); lexically and syntactically flexible.
travel guide	"...here's a quick travel guide full of things you should know about Guadeloupe." Literal, compositional, and flexible (e.g., tourist guide, trip planner).
wedding anniversary	"...in honor of their 25th wedding anniversary." Literal and culturally conventional. Allows modifiers (e.g., silver wedding anniversary), but rephrasing is rare. Mildly flexible.

Table 13: Examples of Rigid/Fixed and Flexible Multi-word Expressions (MWEs) with explanations.

MWE	MB4 Errors	MB5 Errors
panda car	2 FN	8 FN
bow tie	2 FN	1 FN
chain reaction	6 FP	2 FP
flower child	none	3 FN
health check	2 FN	2 FN

Table 14: Error comparison on rigid/fixed MWEs between MB4 and MB5.

MWE	MB4 Errors	MB5 Errors
panda car	2 FN	8 FN
bow tie	2 FN	1 FN
chain reaction	6 FP	2 FP
flower child	none	3 FN
health check	2 FN	2 FN

Table 15: Error comparison on rigid/fixed MWEs between MB4 and MB5.

Language	MWE	Sentence	Label
English	old hat	Serve our favorite bourbon whiskeys in an **old hat** and we'd still probably take a sip	1
English	old hat	But not all of the accouterments of power are **old hat** for the president.	0
Portuguese	força bruta	**Força bruta** continua sendo uma das bombas mais fortes do mundo.	1
Portuguese	força bruta	Gardner é conhecido por ser impulsivo e usar os poderes com grande impacto, de forma instintiva, com **força bruta**.	0

Table 16: Example Sentences from the dataset (Tayyar Madabushi et al., 2022)

A Appendix D

See the detailed results tables and further Tables below

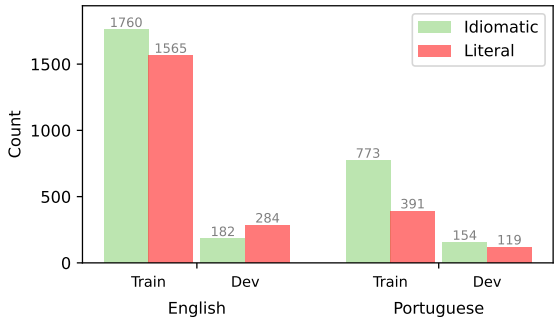

Figure 4: Class Distribution

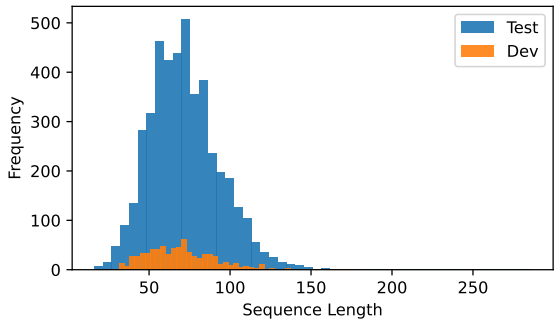

Figure 5: Sequence Length

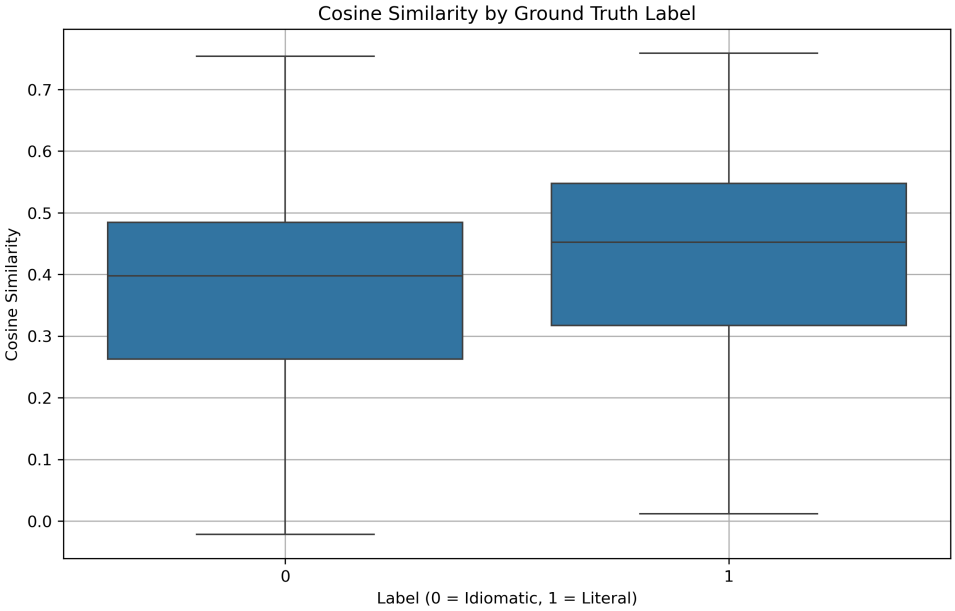

Figure 6: Cosine Similarity by Ground Truth Label.

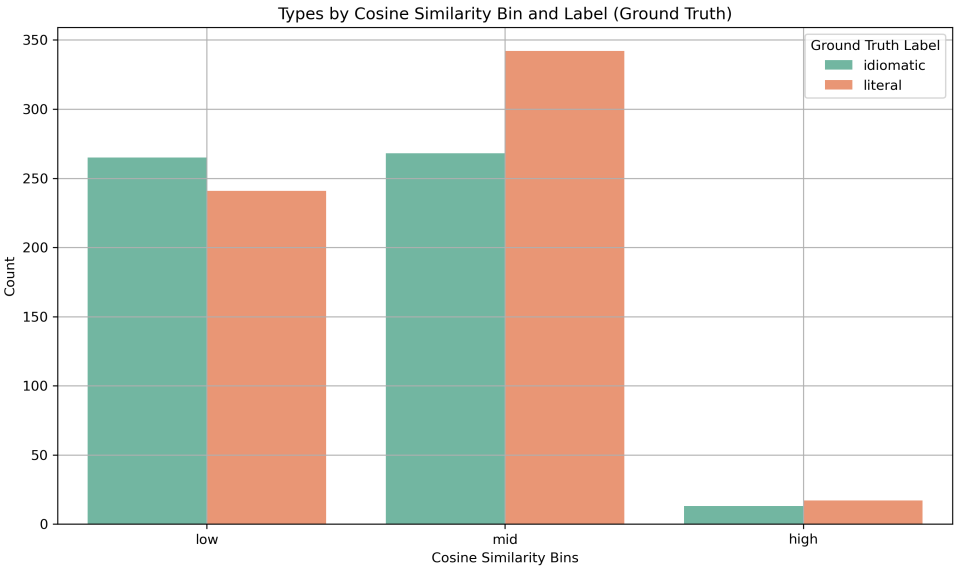

Figure 7: Cosine Similarity Clustered.

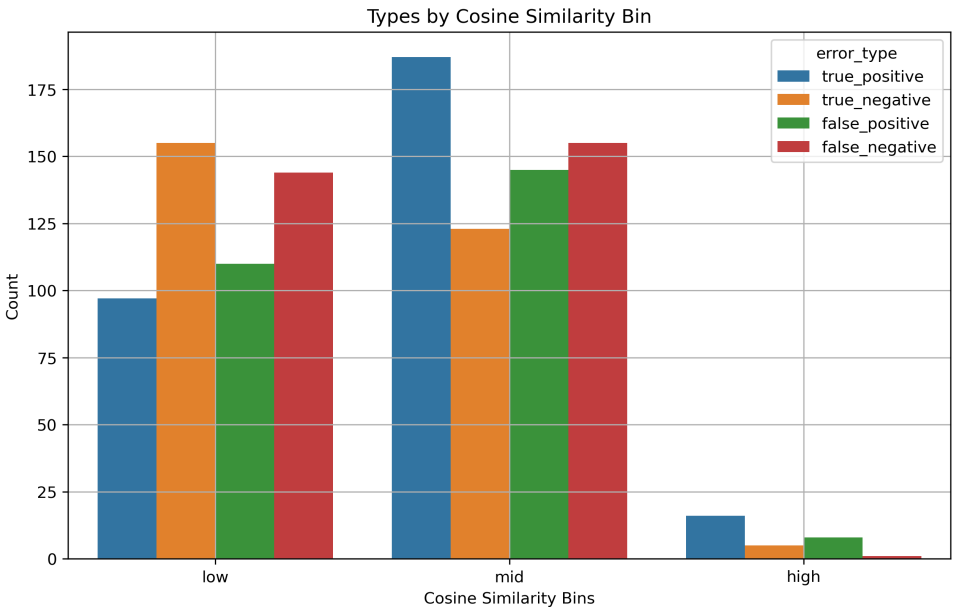

Figure 8: Cosine Similarity: Clustered and binned.